

Constructing a Pose Map

Constructing a Pose Map

1. Course Content
2. Building Grid Maps

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1. Use slam_toolbox to build grid maps and pose maps

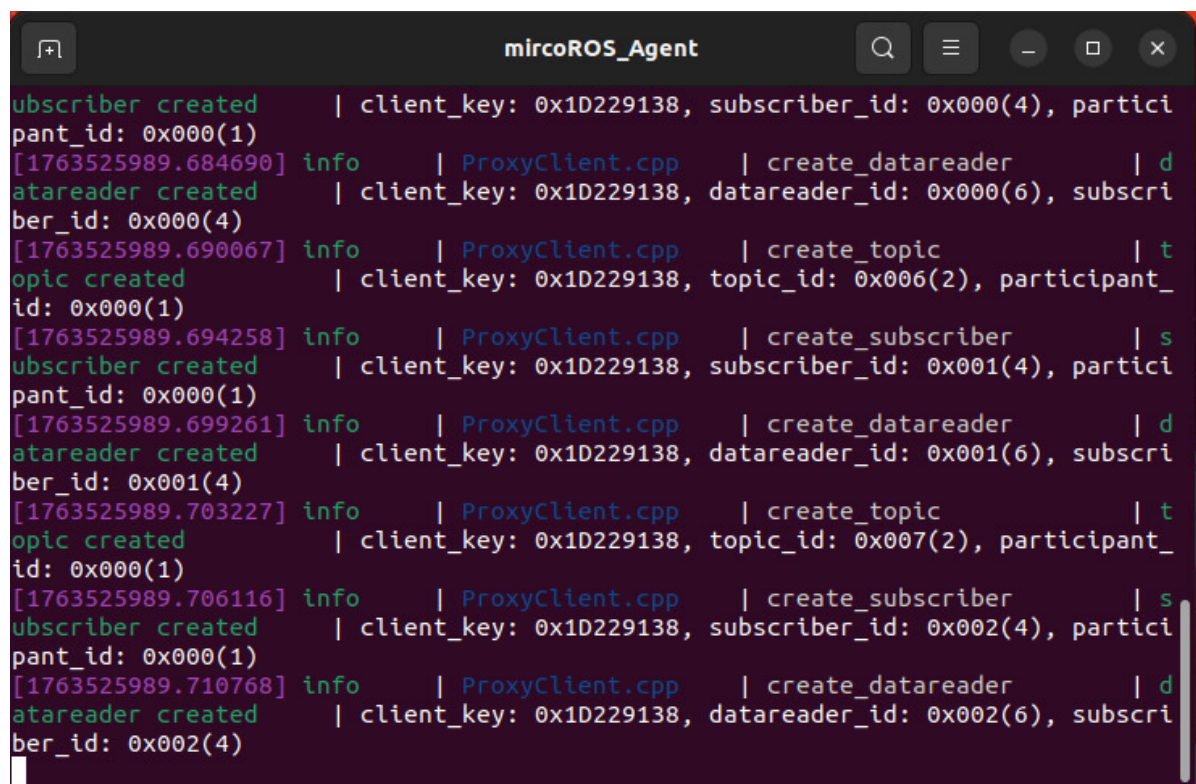
[!NOTE]

- Before marking the road network .geojson file, you need at least one grid map
- Pose maps are used to enable fast relocalization when marking waypoints and during navigation, improving global positioning accuracy and waypoint marking accuracy.

2. Building Grid Maps

- If the chassis microROS proxy communication is not started, you need to start the proxy first. If it is already started, there is no need to start it again.

```
start_agent
```



```
subscriber created | client_key: 0x1D229138, subscriber_id: 0x000(4), partici
pant_id: 0x000(1)
[1763525989.684690] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x1D229138, datareader_id: 0x000(6), subscri
ber_id: 0x000(4)
[1763525989.690067] info | ProxyClient.cpp | create_topic | t
opic created | client_key: 0x1D229138, topic_id: 0x006(2), participant_
id: 0x000(1)
[1763525989.694258] info | ProxyClient.cpp | create_subscriber | s
ubscriber created | client_key: 0x1D229138, subscriber_id: 0x001(4), partici
pant_id: 0x000(1)
[1763525989.699261] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x1D229138, datareader_id: 0x001(6), subscri
ber_id: 0x001(4)
[1763525989.703227] info | ProxyClient.cpp | create_topic | t
opic created | client_key: 0x1D229138, topic_id: 0x007(2), participant_
id: 0x000(1)
[1763525989.706116] info | ProxyClient.cpp | create_subscriber | s
ubscriber created | client_key: 0x1D229138, subscriber_id: 0x002(4), partici
pant_id: 0x000(1)
[1763525989.710768] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x1D229138, datareader_id: 0x002(6), subscri
ber_id: 0x002(4)
```

- Enter commands in the terminal

For Raspberry Pi 5 users, if the ROS container is not started, you need to start the container first,

```
bringup_m3
```

Open a terminal inside the container,

```
exec_m3
```

(ORIN mainboard)

```
slam
```

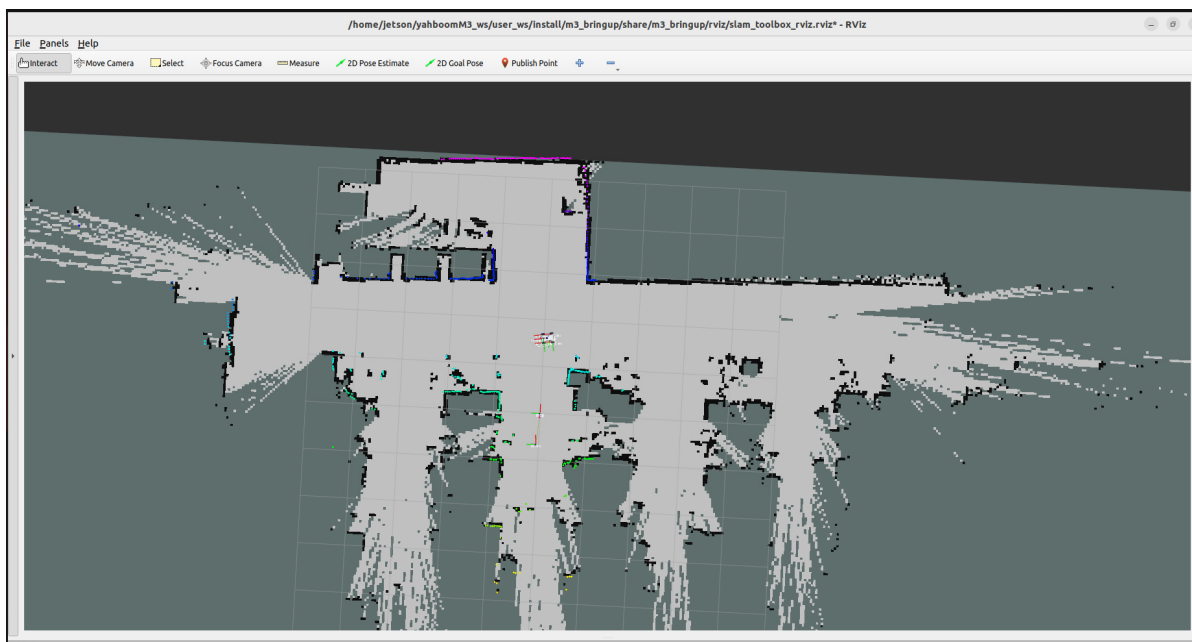
(PI5 & RDK mainboards) Run step by step, with visualization started on the accompanying virtual machine `Rosmaster-M3-Ubuntu22.04`,

```
#Virtual machine side
rviz2 -d
/home/yahboom/yahboomM3_ws/user_ws/src/m3_bringup/rviz/slam_toolbox_rviz.rviz
#Host side
ros2 launch m3_bringup slam_toolbox.launch.py gui:=False
```

- Start the keyboard control node (you can also use a gamepad for control)

```
keyboard_control
```

- Control the vehicle to move for mapping



- Save the grid map

```
ros2 launch m3_bringup save_map.launch.py map_name:=map
```

[!TIP]

- `map_name` is used to specify the **map name** of the saved grid map. If no startup parameter is added, the default map name is `map`, and all map files are saved in the `~/map` directory.

The terminal prompt `Map saved successfully` indicates that the map was saved successfully

```
jetson@yahboom: ~ 82x22
[map_saver_cli-1] map_saver lifecycle node launched.
[map_saver_cli-1] Waiting on external lifecycle transitions to activate
[map_saver_cli-1] See https://design.ros2.org/articles/node_lifecycle.html f
or more information.
[map_saver_cli-1] [INFO] [1763532975.329685271] [map_saver]: Creating
[map_saver_cli-1] [INFO] [1763532975.330069728] [map_saver]: Configuring
[map_saver_cli-1] [INFO] [1763532975.341867665] [map_saver]: Saving map from 'map'
topic to '/home/jetson/map/map' file
[map_saver_cli-1] [WARN] [1763532975.359932772] [map_io]: Image format unspecified
. Setting it to: pgm
[map_saver_cli-1] [INFO] [1763532975.361316136] [map_io]: Received a 224 X 598 map
@ 0.05 m/pix
[map_saver_cli-1] [INFO] [1763532975.439369065] [map_io]: Writing map occupancy da
ta to /home/jetson/map/map.pgm
[map_saver_cli-1] [INFO] [1763532975.443013992] [map_io]: Writing map metadata to
/home/jetson/map/map.yaml
[map_saver_cli-1] [INFO] [1763532975.444847063] [map_io]: Map saved
[map_saver_cli-1] [INFO] [1763532975.446514786] [map_saver]: Map saved successfull
y
[map_saver_cli-1] [INFO] [1763532975.449017438] [map_saver]: Destroying
[INFO] [map_saver_cli-1]: process has finished cleanly [pid 50218]
jetson@yahboom:~$
```

- Save slam_toolbox pose map

```
ros2 launch m3_bringup save_map.launch.py map_name:=map map_type:=posegraph
```

[!TIP]

- `map_type` saves the map type, default is `gridmap` to save grid map, select `posegraph` to save `slam_toolbox` pose map

Here the terminal prompt `slam_toolbox.srv.SerializePoseGraph_Response(result=0)` indicates that the pose map has been saved

```
jetson@yahboom: ~ 82x22
jetson@yahboom:~$ ros2 launch m3_bringup save_map.launch.py map_name:=/home/jetson
/map/map map_type:=posegraph
[INFO] [launch]: All log files can be found below /home/jetson/.ros/log/2025-11-19
-14-19-41-297598-yahboom-66778
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [ros2-1]: process started with pid [67023]
[ros2-1] requester: making request: slam_toolbox.srv.SerializePoseGraph_Request(fi
lename='/home/jetson/map/map')
[ros2-1]
[ros2-1] response:
[ros2-1] slam_toolbox.srv.SerializePoseGraph_Response(result=0)
[ros2-1]
[INFO] [ros2-1]: process has finished cleanly [pid 67023]
jetson@yahboom:~$
```

- Map files are saved in the `~/map` path by default. You can specify the map name through the `map_name` parameter when saving the map

